

Project Design Phase-II: Solution Requirements (Functional & Non-functional)

Project Overview

- **Project Name:** Heart Disease Analysis[cite: 27]
- **Date:** 8 July 2026[cite: 27]
- **Maximum Marks:** 4 Marks[cite: 27]
- **Team Leader:** Mounika Sanapathi[cite: 27]
- **Team Members:** Thakasi Balaji, Kale Sai Praneeth, Navyasri Gonnuri, Shanthi Kumari Challa[cite: 27]

Functional Requirements

The following are the functional requirements of the proposed clinical analytics solution.
[cite: 27]

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Dashboard Visualization[cite: 27]	• Display an interactive Tableau dashboard containing vital patient metrics.[cite: 27] • Display clinical KPI summary cards for total patient count, high-risk case distributions, and average patient survival rates.[cite: 27]
FR-2	Demographic & Risk Analysis[cite: 27]	• Analyze heart disease prevalence and distribution across distinct age groups and genders.[cite: 27] • Compare patient risk factor trends and demographic distributions across different regions. [cite: 27]

FR-3	Biometric Factor Analysis[cite: 27]	• Display patient distributions categorized by primary chest pain and angina types.[cite: 27] • Show multi-variable risk correlations spanning cholesterol levels, resting blood pressure, and fasting blood sugar markers.[cite: 27]
FR-4	Diagnostic Test Analysis[cite: 27]	• Analyze medical metrics such as resting electrocardiogram (ECG) results across patient cohorts.[cite: 27] • Compare diagnostic trends and diagnostic outcomes across distinct patient risk classifications. [cite: 27]
FR-5	Lifestyle & Stress Performance[cite: 27]	• Display continuous trends analyzing maximum heart rate achieved during exercise stress tests.[cite: 27] • Compare the impact of external lifestyle behaviors (smoking, physical activity) across age groups and clinical segments.[cite: 27]
FR-6	Regional Health Analysis[cite: 27]	• Analyze clinical case volumes and cardiovascular disease rates by geographical region.[cite: 27] • Analyze the overall distribution of critical vs. stable patient metrics across regional clinical centers.[cite: 27]
FR-7	Interactive Diagnostic Filters[cite: 27]	• Apply global medical dashboard filters by region, chest pain type, age group, and lifestyle indicators. [cite: 27] • Enable direct clinical drill-down capabilities within individual visualizations.[cite: 27]
FR-8	Tableau Medical Story[cite: 27]	• Display a guided analytical story mapping epidemiological and patient risk trajectories.[cite: 27] • Present key clinical discoveries and population health trends in a narrative format.[cite: 27]
FR-9	Flask Clinical Website[cite: 27]	• Embed the interactive Tableau diagnostic dashboard securely within the central Flask web application portals.[cite: 27] • Embed the guided Tableau medical story layout directly within the Flask web platform.[cite: 27]

FR-10	Clinical Reporting[cite: 27]	• Present summarized data-driven insights and clinical summaries for healthcare executives.[cite: 27] • Support high-level decision-making for medical directors, chief cardiologists, and hospital administrators.[cite: 27]
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Non-functional Requirements

The following are the non-functional requirements specified for the clinical analytics platform.[cite: 27]

FR No.	Non-Functional Requirement	Description
NFR-1	Usability[cite: 27]	The patient dashboard should be easy to use, providing clear navigation menus and intuitive interactive visualizations tailored for medical professionals and clinical analysts.[cite: 27]
NFR-2	Security[cite: 27]	The application should securely present healthcare metrics through the Flask web interface without exposing raw, sensitive, or identifiable patient Electronic Health Records (EHR).[cite: 27]
NFR-3	Reliability[cite: 27]	The dashboard must consistently produce precise, dependable, and clinically accurate analytical results generated from the underlying patient health dataset. [cite: 27]
NFR-4	Performance[cite: 27]	Medical dashboard views must load efficiently; charts, demographic filters, and risk cross-filtering variables should respond smoothly during user interaction.[cite: 27]
NFR-5	Availability[cite: 27]	The Tableau dashboards and hosting Flask clinical website must remain fully operational and accessible to hospital administrators whenever urgent business or health insights are required.[cite: 27]

NFR-6	Scalability[cite: 27]	The solution must successfully accommodate larger health record datasets and integrate additional sub-dashboards as clinical research parameters expand in the future.[cite: 27]
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Prepared as part of the Heart Disease Analysis and Patient Risk Trends capstone project.[cite: 27]